PDF HTML5 PAGE FLIP

Next-Gen 3D Digital Flipbook Engine

Open-Source Synthetic Showcase Document

This document is purely generated for rendering and performance testing.

Contains complete navigation outlines and table of contents.

Built with Vite, TypeScript, PDF.js, and HTML5 Canvas.

60 FPS HARDWARE ACCEL

1. Architecture & Rendering Engine

HIGH PERFORMANCE CLIENT-SIDE PDF RENDERING

Off-Main-Thread Processing with Web Workers

Mozilla PDF.js decodes PDF binary streams inside a dedicated worker, preventing UI thread freezes during intensive document rasterization.

Rendering Pipeline Highlights:

- HiDPI & Retina display automatic viewport devicePixelRatio scaling
- Lazy virtualization rendering: only visible pages render to canvas
- Active RenderTask cancellation on rapid user page flipping
- 3D curved spine shadow and realistic paper curl simulation

2. Performance Benchmarks

COMPUTATIONAL EFFICIENCY ON DESKTOP AND MOBILE

Feature / Capability	Legacy Turn.js	Page Flip Pro v2
Touch & Swipe Support	Limited (jQuery)	**Native Touch Events**
Animation Easing & FPS	~30-45 FPS	**Solid 60 FPS (GPU)**
Audio Synthesis	Audio Element	**Web Audio API**

3. React Component Usage

ZERO-CONFIGURATION MODERN FRONTEND WORKFLOW

```
// Install package from npm
npm install pdf-html5-page-flip

// Import in your Next.js / React application
import { PdfFlipbook } from 'pdf-html5-page-flip/react';
import 'pdf-html5-page-flip/style.css';

export default function Reader() {
    return <PdfFlipbook src="/my-book.pdf" theme="light" />;
}
```

4. Interactive Controls & Features

INTUITIVE NAVIGATION AND READING TOOLS

Keyboard Shortcuts

Left / Right Arrows : Prev / Next

Home / End : First / Last Page

F Key : Toggle Fullscreen

Sidebar & Tools

Outline : Hierarchical TOC

Search : Full-text query

Thumbnails : Visual preview

Ready to Embed in Your App

Deploy with zero dependencies on modern browser standards.

License: Apache License 2.0

Repository: github.com/iamapinan/pdf-html5-page-flip

Live Demo: https://flipbook.apps.gracer.ai